LAGOS BUSINESS SCHOOL · MEMBA 14 · EEGB · PART I

SESSION 2

Demand, Supply & Elasticity with Behavioural Insights

From markets, to macro, to decisions that matter — in Nigeria and the world.



❐ Reading: Note on Microeconomics for Managers · Facilitator: Bongo Adi

RECAP

Session 1 in Three Ideas

Think like an economist

Incentives, opportunity cost and decisions at the margin frame every business choice.

Nominal vs real

Take inflation and the exchange rate out of the numbers before celebrating growth.

Read the data

Know your sources; watch for rebasing, multiple rates and correlations that are not causes.

TODAY'S PLAN

Session 2 at a Glance — 75 Minutes

How prices are set, how customers respond, and how to price through a cost shock.



?

Recap & Hook

Flour is up 40%. What do we charge?

⚖

Demand & Supply

Why prices move, and who bears a cost shock

↗

Elasticity

Own, cross and income — and why the naive number lies

✦

Behaviour & Pricing

Reference prices, promotions, dynamic pricing, pass-through

BY THE END OF TODAY

Session 2 Learning Objectives

You will be able to:

01

Read a Market

Use demand and supply to explain why a price moved, and predict who bears a cost increase.

02

Measure Elasticity

Estimate own-price, cross-price and income elasticities from sales data — holding other things constant.

03

Price Under Uncertainty

Decide how much of a cost shock to pass on, using elasticity, rivals' behaviour and customers' reference prices.

THE HOOK

Flour Is Up 40%. What Do We Charge?

“Our flour bill has jumped 40%. Just add it to the price — customers have no choice, everybody eats noodles.”

Before signing off the new price list, a good manager asks three questions:

↗ How sensitive are buyers?

If a 10% price rise loses 13% of volume, 'just pass it on' shrinks revenue.

☺ What will rivals do?

If competitors face the same shock and also raise prices, we lose far fewer customers.

✦ What price feels 'fair'?

Shoppers compare with the price they are used to. Big jumps hurt more than the arithmetic suggests.

NIGERIAN BUSINESS CASE

Pricing Noodles Through a Flour Shock

The situation

A Nigerian noodles maker sells a single-serve pack in three regions — Lagos, the North and the South-East. Shelf prices have more than tripled since 2022 as costs and the exchange rate moved.

The decision question

Flour, about **45% of unit cost**, has just risen **40%.** How much of the extra cost should be **passed into price**, and how fast?

The data we have

fmcg_price_volume.csv — 209 weeks × 3 regions: shelf price, competitor price, promotion weeks, an income index and packs sold. *Stylised teaching data.*

What a good answer looks like

A pricing recommendation that uses **measured elasticities**, a view on **rivals' response** and the risk of breaking customers' **reference price.**

PART

01

Demand, Supply and Price

The two forces behind every price tag.

What Moves Demand, and What Moves Supply

Demand: willingness to buy

Slopes down — higher price, fewer packs

- Price of the product (a move along the curve)
- Income, prices of substitutes and complements
- Tastes, population, expectations of future prices
- These shift the whole curve

Supply: willingness to sell

Slopes up — higher price, more packs

- Price of the product (a move along the curve)
- Input costs: flour, diesel, packaging, FX
- Technology, number of sellers, taxes and levies
- These shift the whole curve

*A change in **price** moves you **along** a curve. A change in **anything else shifts** the curve. Most management mistakes come from mixing the two.*

Who Pays When Flour Gets Dearer?

A cost shock shifts supply up: the price rises by less than the cost, and volume falls

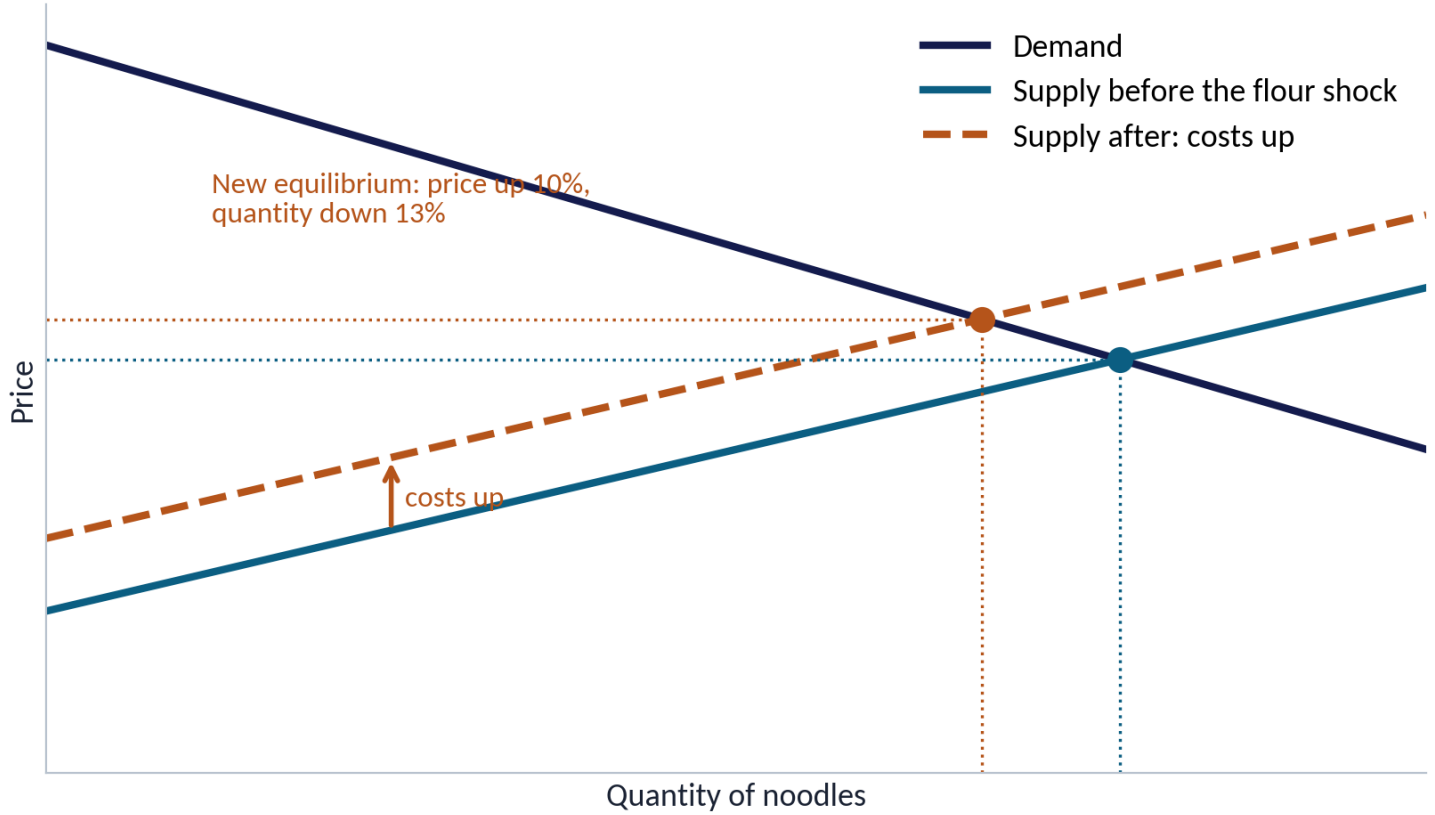


Fig S02-F01 · replicate: s02_figures.R / .py

WHAT TO SEE

- Higher flour costs shift the supply curve **up**: firms need a higher price to supply the same quantity.
- The new equilibrium has a **higher price (+10%)** and a **lower quantity (−13%)** in this example.
- The price rises by **less than the cost**: part of the shock is paid by customers, part by producers.
- How it splits depends on elasticity: the less sensitive the buyers, the more they pay.

NIGERIAN EXAMPLES

Shifts You Have Lived Through

Petrol subsidy removal (2023)

A supply-side cost shock for almost every business: transport, generators and distribution all became dearer at once.

Naira depreciation

Imported wheat, packaging and machinery cost more naira: supply shifts up across manufacturing.

Falling real incomes

When wages lag inflation, demand for non-essentials shifts **down** — consumers trade down to smaller packs and cheaper brands.

New sellers online

E-commerce and social-media sellers add supply and make prices easier to compare — demand facing each seller becomes more elastic.

PART

02

Elasticity: How Much Do Customers React?

One number that should sit beside every price decision.

THE KEY NUMBER

Own-Price Elasticity of Demand

IN PLAIN ENGLISH

Elasticity tells you **by what percentage volume changes when price changes by 1%**. If the number is bigger than 1 (ignoring the minus sign), demand is **elastic**: customers react strongly. Smaller than 1 is **inelastic**.

EXAMPLE

Our noodles: elasticity ≈ **−1.47**. A 10% price rise loses about **13%** of volume when rivals hold their prices.

THE FORMULA

$$\text{elasticity} = \frac{\%\ \text{change in quantity}}{\%\ \text{change in price}}$$

WHY A MANAGER CARES

Elastic demand means a price rise **reduces revenue**; inelastic demand means it **raises revenue**. Knowing which one you face is the start of every pricing decision.

Prices Tripled — Volumes Did Not Collapse

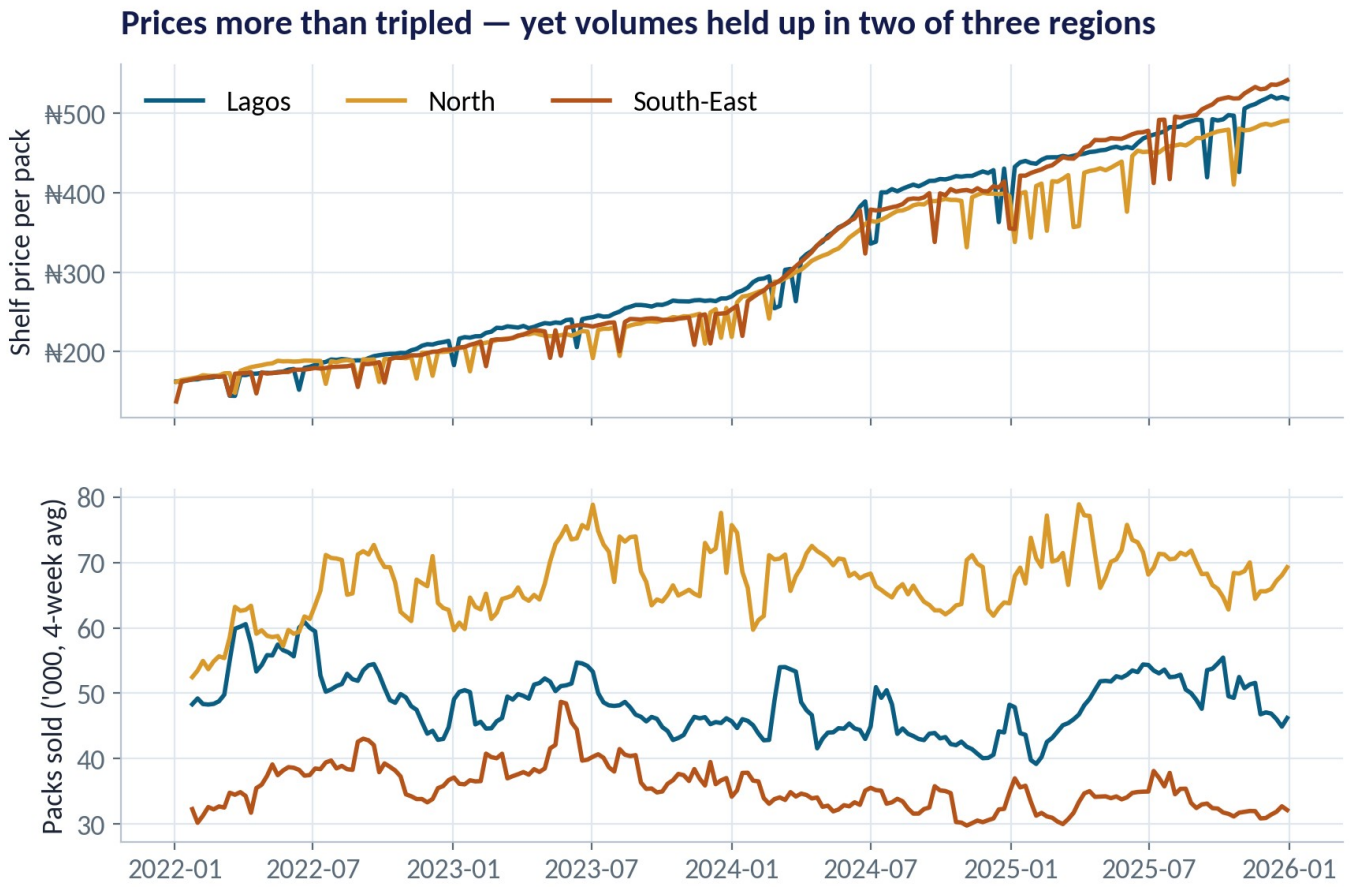


Fig S02-F02 · replicate: s02_figures.R / .py

WHAT TO SEE

- Lagos shelf price went from about **~~₦162~~ to ~~₦518~~** a pack between 2022 and 2025.
- Volumes rose in the North and Lagos, and fell in the South-East.
- Looks like customers do not care about price. **Do not believe it yet.**
- Incomes, competitors' prices and the general price level all moved at the same time.

Inflation Hides the Elasticity

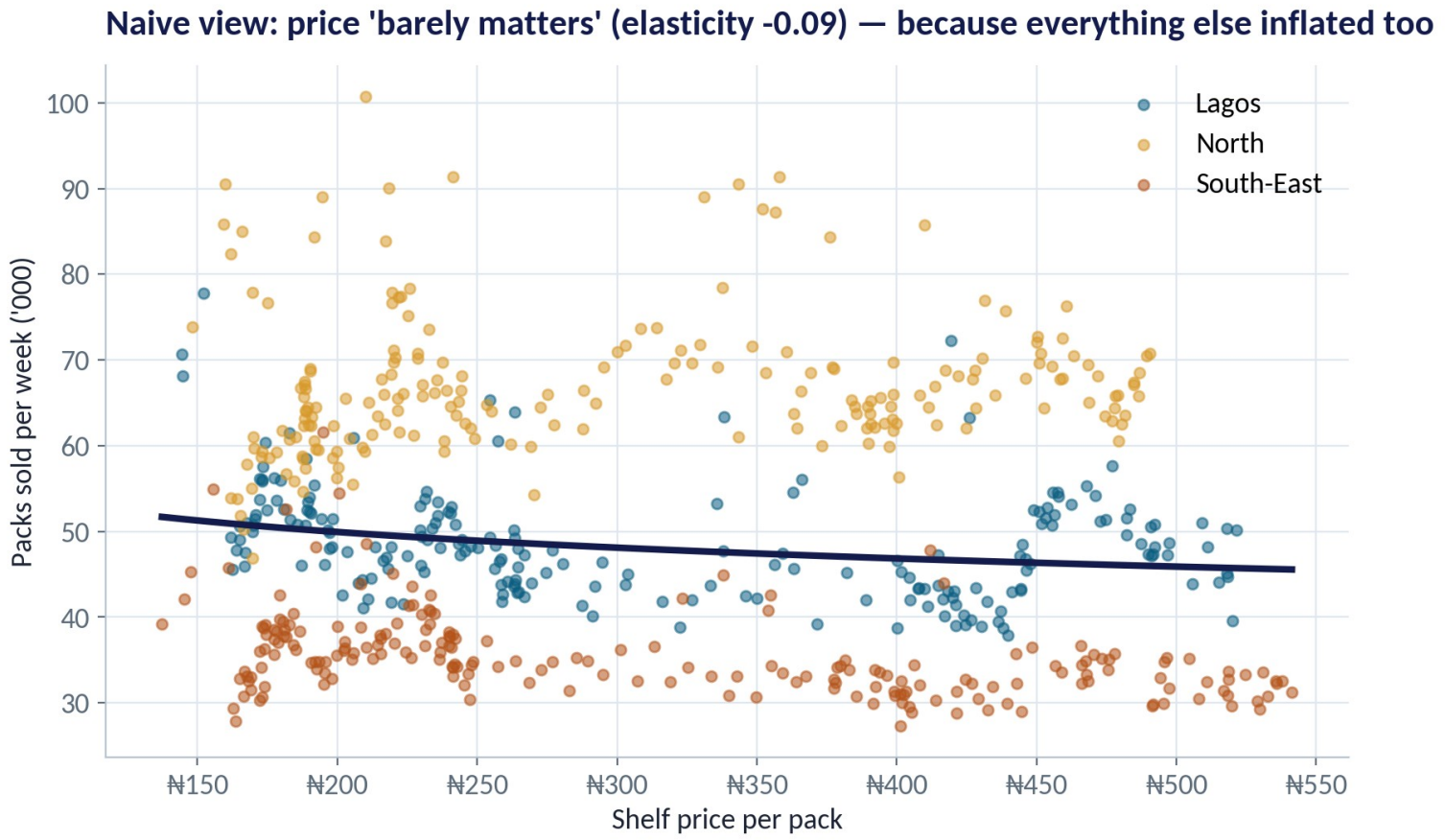


Fig S02-F03 · replicate: s02_figures.R / .py

WHAT TO SEE

- Plotting units against shelf price gives an elasticity of just **−0.09** — 'price barely matters'.
- But **every** price was rising: competitors' prices, wages, the whole price level.
- What matters to a shopper is our price **relative** to the alternatives and to their income.
- To see the true response we must **hold the other drivers constant**.

What Customers Really Respond To

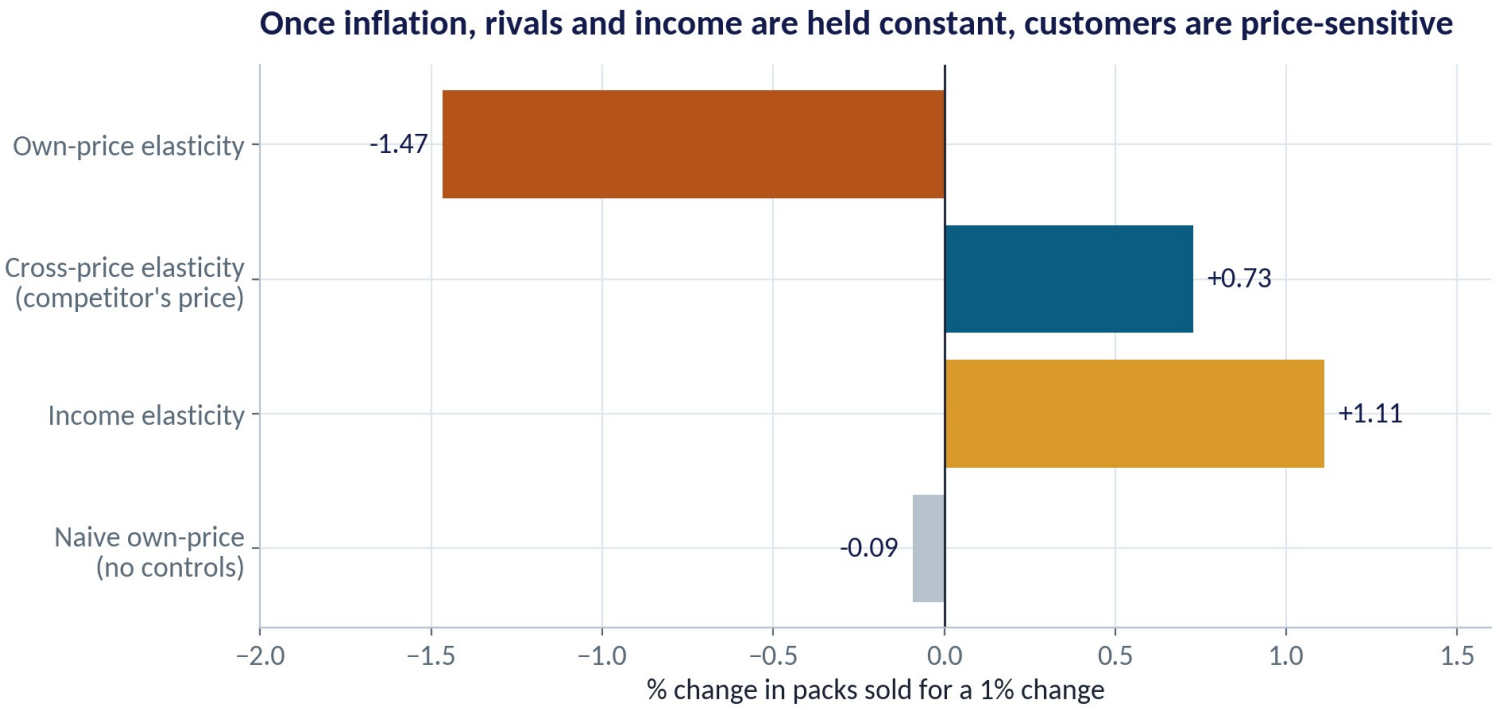


Fig S02-F04 · replicate: s02_figures.R / .py

WHAT TO SEE

- **Own price −1.47**: a 1% rise in our price, all else equal, cuts volume by about 1.5%.
- **Cross price +0.73**: when a rival raises its price by 1%, we gain about 0.7% volume — they are close substitutes.
- **Income +1.11**: as customers' incomes rise, they buy more.
- Method: one regression of log(units) on log(own price), log(rival price), log(income), a time trend and region.

Three Elasticities Every Manager Should Know

Elasticity	Question it answers	Our estimate	Business use
Own-price	If we raise our price 1%, what happens to our volume?	−1.47	Price setting, revenue forecasts
Cross-price	If a rival raises its price 1%, what happens to our volume?	+0.73	Is the rival a close substitute? Will they follow?
Income	If customers' real income rises 1%, what happens to our volume?	+1.11	Planning through recessions and booms; trading down
Advertising	If we spend 1% more on ads, what happens to volume?	not estimated	Marketing budgets (usually small, 0.05–0.2)

*Positive cross-price elasticity = **substitutes** (rival noodles). Negative = **complements** (noodles and seasoning).*

When Does a Price Rise Pay?

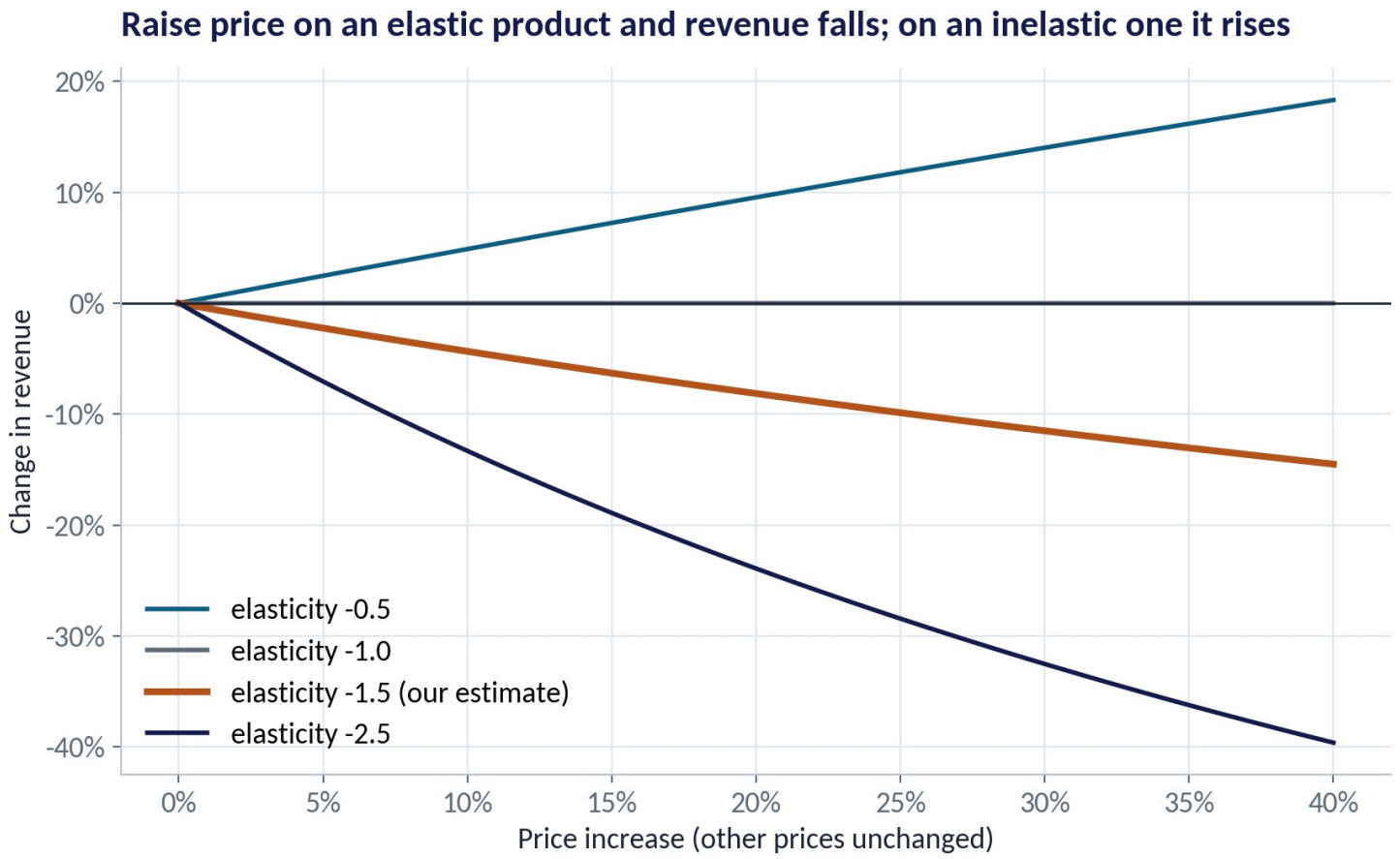


Fig S02-F05 · replicate: s02_figures.R / .py

WHAT TO SEE

- With elasticity −0.5, a 20% price rise **raises** revenue.
- At our estimate (−1.47), a 20% rise **cuts** revenue by about **8%**.
- But revenue is not profit: if the price rise covers higher costs, it can still protect margin.
- Elasticity is lower when **all** firms raise prices together — the market is less elastic than a single brand.

CONCEPT CHECK

Quick Check: What Would You Say?

?

A competitor cuts its noodle price by 10%. Our cross-price elasticity is +0.73. Holding our price, what happens to our volume?

- **A** Rises about 7%
- **B** Falls about 7%
- **C** Falls about 15%
- **D** No change — cross-price only matters for complements

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

PART

03

Behaviour, Promotions and Pass-Through

Customers are not calculators.

Reference Prices and Loss Aversion

IN PLAIN ENGLISH

Shoppers judge a price against the price they are **used to** — their **reference price**. Paying more than usual feels like a **loss**, and people weigh losses about twice as heavily as equal gains.

EXAMPLE

Weeks priced **more than 10% above** the usual price sold about **7.7%** fewer packs than the elasticity alone predicts.

THE FORMULA

pain of a ₦50 rise > pleasure of a ₦50 cut

WHY A MANAGER CARES

Several small, explained increases usually cost less volume than one big jump. Pack-size changes ('shrinkflation'), clear communication and timing all manage the reference price.

Big Jumps Hurt More Than the Arithmetic Says

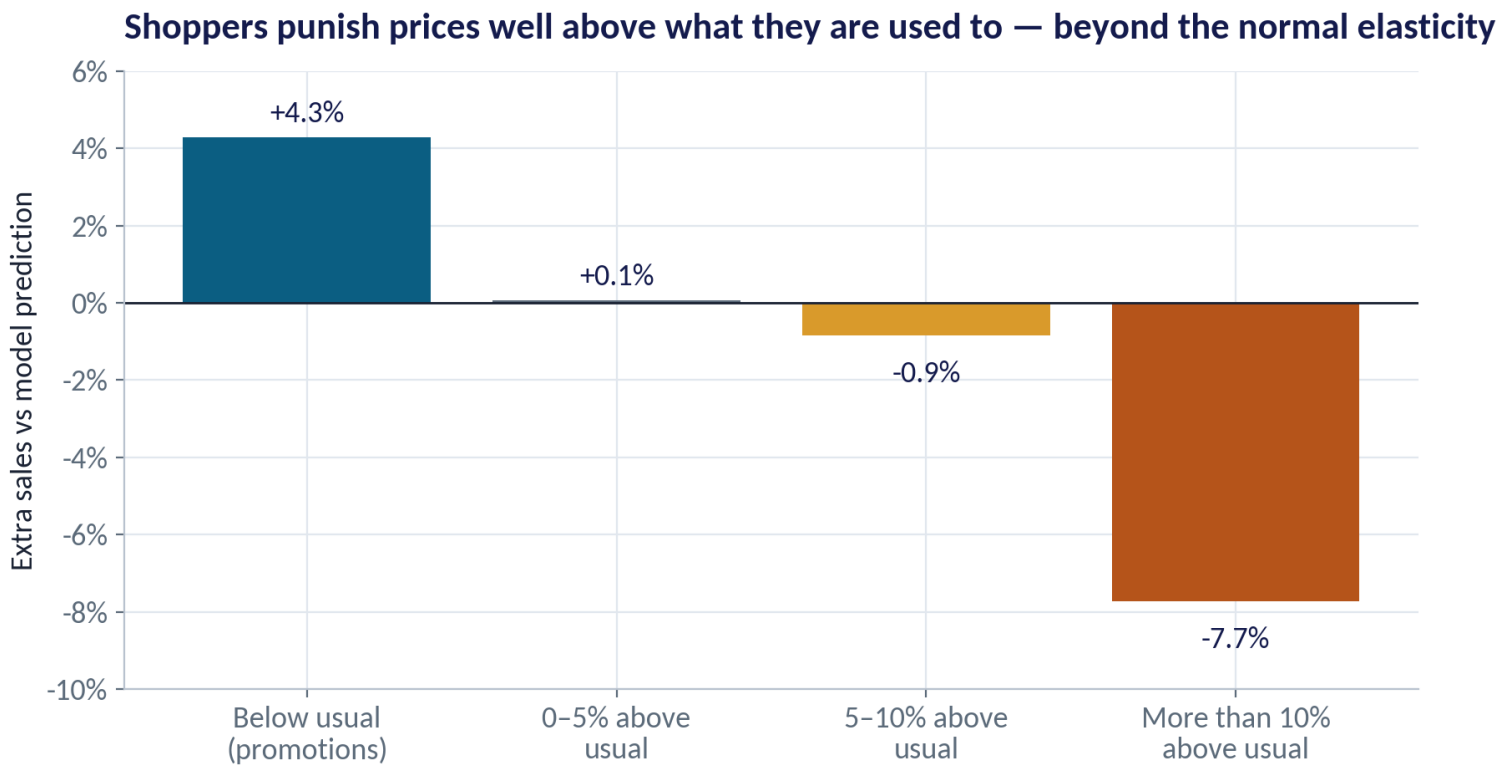


Fig S02-F06 · replicate: s02_figures.R / .py

WHAT TO SEE

- We compared actual sales with what the elasticity model predicts.
- Promotion weeks (price **below** usual) sold about **4%** more than predicted.
- Prices up to 5% above usual: no extra penalty. **More than 10% above: −7.7%.**
- The penalty fades as the new price becomes the new 'normal' — reference prices adapt over weeks.

What a 15% Discount Buys

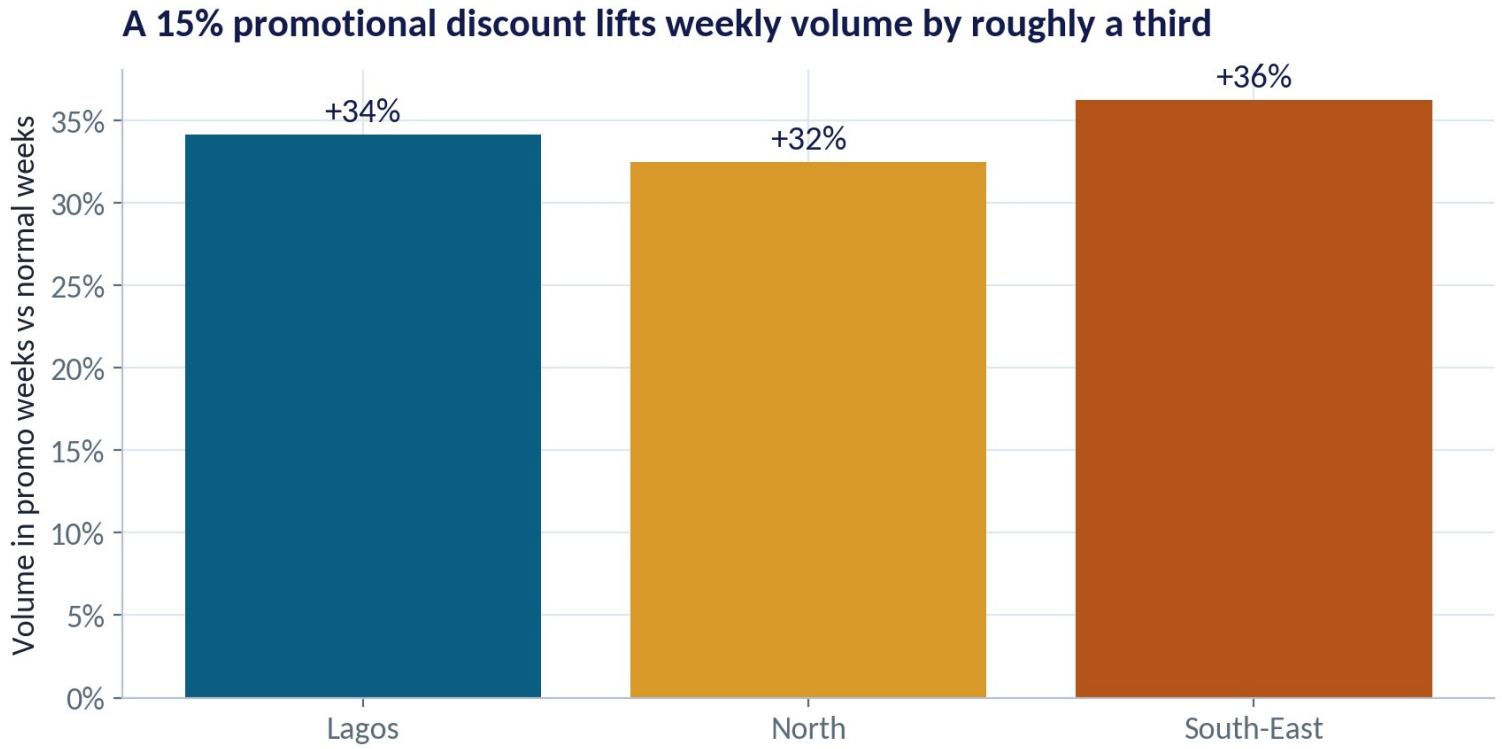


Fig S02-F07 · replicate: s02_figures.R / .py

WHAT TO SEE

- Promotion weeks (15% off) sold **32–36%** more packs than normal weeks.
- Part of that is genuinely new demand; part is **pantry loading** — customers buying earlier, not more.
- Measure the weeks **after** a promotion too: a dip means you borrowed sales from the future.
- Promotions also reset reference prices downward if they run too often.

Dynamic Pricing — Power and Limits

IN PLAIN ENGLISH

Dynamic pricing changes prices often, using data on demand, stock, time and competitors. Ride-hailing surge pricing and airline fares are examples. AI makes it cheap to estimate elasticities by product, store and hour.

EXAMPLE

A ride-hailing app raises fares in Lagos rush hour and during heavy rain, when many riders compete for few drivers.

THE FORMULA

price ↑ when demand is high or capacity is scarce

WHY A MANAGER CARES

It can raise revenue and balance supply and demand, but it can also feel **unfair** and invite regulators. Rules of thumb: be transparent, cap surges on essentials, and never personalise prices in ways customers would object to.

How Much of the Flour Shock to Pass On?

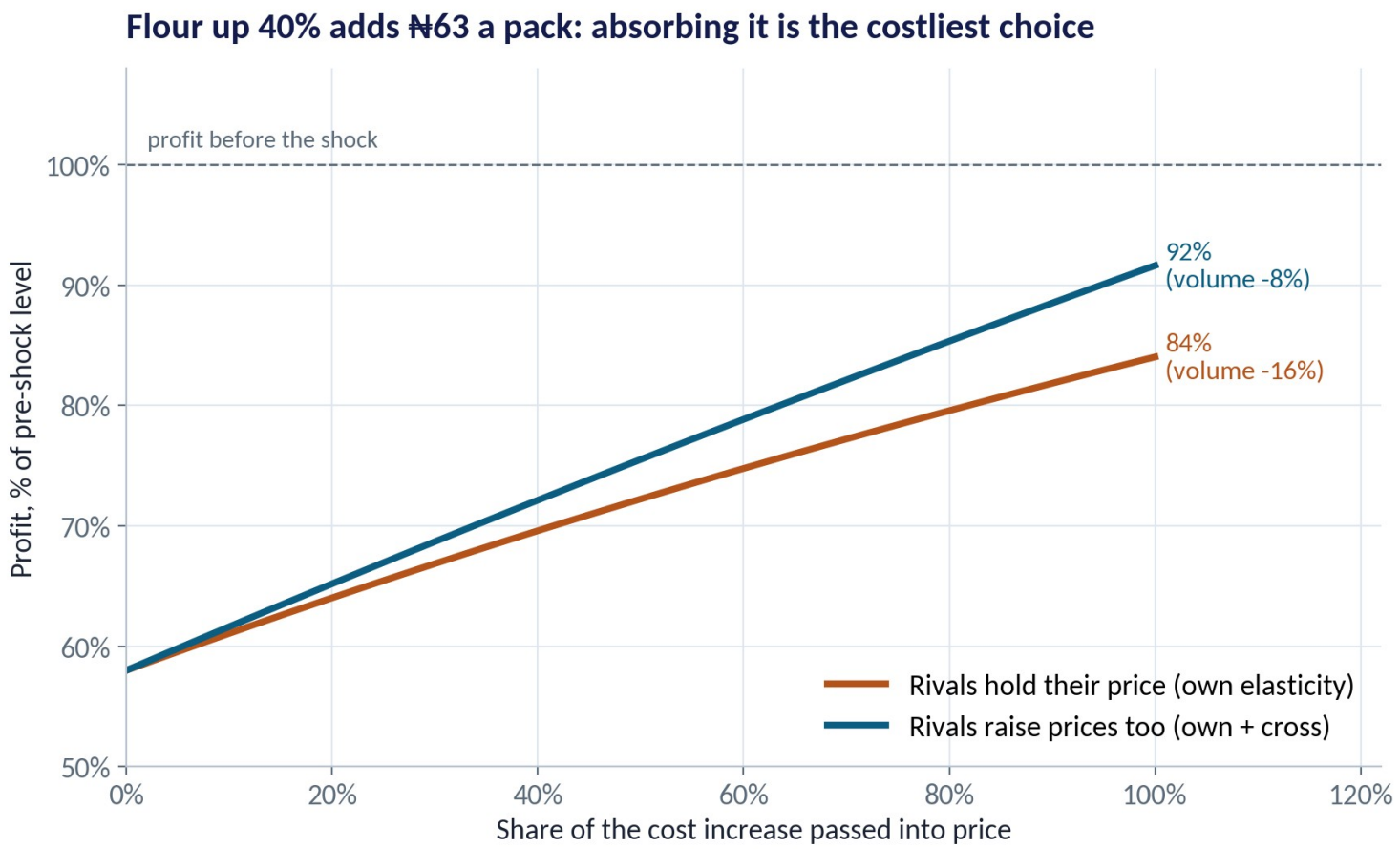


Fig S02-F08 · replicate: s02_figures.R / .py

WHAT TO SEE

- The shock adds **₦63** to a pack that sells for ₦500 and costs ₦350 to make.
- **Absorb it all**: volume holds, but profit falls to **58%** of what it was.
- **Pass it all on**, rivals holding: volume −16%, profit back to **84%**. If rivals follow: volume −8%, profit **92%**.
- Since all millers face the same flour price, rivals are **likely to follow** — so passing on most of it, in steps, is the stronger option.

WHERE MANAGERS GO WRONG

Common Pricing Mistakes

THE TRAP	THE BETTER HABIT
✗ “Volumes rose while prices rose, so demand is not price-sensitive.”	✓ Hold **inflation, rivals' prices and income** constant before reading the elasticity.
✗ Using one elasticity for every situation.	✓ Distinguish **our brand alone** (more elastic) from **the whole market moving together** (less elastic).
✗ Passing on a big cost increase in one jump.	✓ Stagger increases and manage the **reference price**: explain, resize, time them.
✗ Judging a promotion by the weeks it ran.	✓ Count the **dip afterwards** and the margin given away.
✗ Treating revenue as the goal.	✓ Decide on **profit** and **market position**, not revenue alone.

Quick Check: What Would You Say?

?

Flour costs rise 40% for every noodles maker in Nigeria. Which statement is most accurate?

A We should absorb the cost; any price rise loses customers to rivals

B Rivals face the same shock, so the relevant elasticity is closer to the market's (less elastic) — passing on most of the cost in steps is likely to protect profit

C Pass on exactly 40% because flour rose 40%

D Price does not matter; the naive elasticity is close to zero

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

HANDS-ON LAB

Lab: How Price-Sensitive Are Our Customers?

YOU WILL NEED

Data
`data/fmcg_price_volume.csv`

R script
`sessions/s02/R/s02_lab.R`

Python script
`sessions/s02/python/s02_lab.py`

⏱ 20 minutes

1. Look at prices and volumes by region from 2022 to 2025.
2. Estimate the **naive elasticity** (units on price only) and explain why it misleads.
3. Estimate own-price, cross-price and income elasticities **holding other things constant**.
4. Compute the volume and revenue effect of 5%, 10% and 20% price rises.
5. Test pass-through of 0%, 50% and 100% of the flour shock — with rivals holding and following.

Deliverable: a 3-sentence pricing recommendation to the MD: how much to pass on, how fast, and what to watch.

WORDS TO OWN

Today's Vocabulary in Plain English

Demand curve	How much buyers want at each price, all else equal.
Supply curve	How much sellers offer at each price, all else equal.
Equilibrium	The price at which quantity demanded equals quantity supplied.
Shift vs move	Price changes move along a curve; other drivers shift it.
Own-price elasticity	% change in quantity for a 1% change in own price.
Cross-price elasticity	% change in our quantity for a 1% change in a rival's price.
Income elasticity	% change in quantity for a 1% change in income.
Reference price	The price a shopper expects, based on recent experience.
Loss aversion	Losses feel worse than equal-sized gains feel good.
Pass-through	Share of a cost change that ends up in the selling price.

KEY TAKEAWAYS

Session 2 in 5 Ideas

01 Prices come from **demand and supply**; a cost shock shifts supply and is **shared** between buyers and sellers.

02 Separate a **move along** a curve from a **shift** of the curve.

03 Measure elasticity **holding other things constant** — inflation hides it.

04 **Own, cross and income** elasticities tell you how volume reacts to you, rivals and the economy.

05 Customers judge prices against a **reference price**: stagger, explain and time increases.



NEXT › Session 3: Forecasting & Market Sizing

Before next session: Read Sahay, Time Series Analysis and Forecasting (BEP8370), the chapters on trend and seasonality. Bring one demand number your company forecasts every year and how far off last year's forecast was.